

ST117 Individual DRAFT Written Report - Phase 2 Task 1 (Woodland Vegetation – Tree Measurements)

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1. u5685279

a)

```
library(dplyr)
library(tidyr)

wd <- read.csv("ECN_VW1.csv", stringsAsFactors = FALSE)

wd_sub <- wd %>%
  filter(FIELDNAME %in% c("DIAMETER", "HEIGHT", "DISTANCE", "NUM_STEMS")) %>%
  select(SITECODE, SYEAR, TREE_SPEC, FIELDNAME, VALUE)

wd_sub$VALUE <- suppressWarnings(as.numeric(wd_sub$VALUE))

wd_wide <- wd_sub %>%
  group_by(SITECODE, SYEAR, TREE_SPEC, FIELDNAME) %>%
  summarise(VALUE = mean(VALUE, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = FIELDNAME, values_from = VALUE)

head(wd_wide, 3)
```

```
## # A tibble: 3 x 7
##   SITECODE SYEAR TREE_SPEC DIAMETER HEIGHT NUM_STEMS DISTANCE
##   <chr>    <int>   <int>   <dbl> <dbl>    <dbl>    <dbl>
## 1 T05      1994     237    19.4  23.6      1        NA
## 2 T05      1994     441      8      7     14.5      NA
## 3 T05      1994    2600    6.88  8.06      1        NA
```

b)

Diameter was used as the predictor and height as the response because the EDA suggested a clear positive relationship between tree size variables, where larger diameters tend to correspond to taller trees. This makes biological sense and fits a simple linear regression framework for modelling structural growth patterns.

```
# Fit model
model <- lm(HEIGHT ~ DIAMETER, data = wd_wide)

# Create coefficient table
coef_table <- as.data.frame(summary(model)$coefficients)
colnames(coef_table) <- c("Estimate", "Std.Error", "t value", "Pr(>|t|)")

# Improve spacing + readability
coef_table <- round(coef_table, 4)
```

```
# Print formatted table
knitr::kable(
  coef_table,
  caption = "Linear regression of HEIGHT on DIAMETER",
  align = "lcccc"
)
```

Table 1: Linear regression of HEIGHT on DIAMETER

	Estimate	Std.Error	t value	Pr(> t)
(Intercept)	8.1013	0.5872	13.7956	0
DIAMETER	0.1130	0.0151	7.4908	0

c)

```
library(dplyr)
library(ggplot2)
library(knitr)

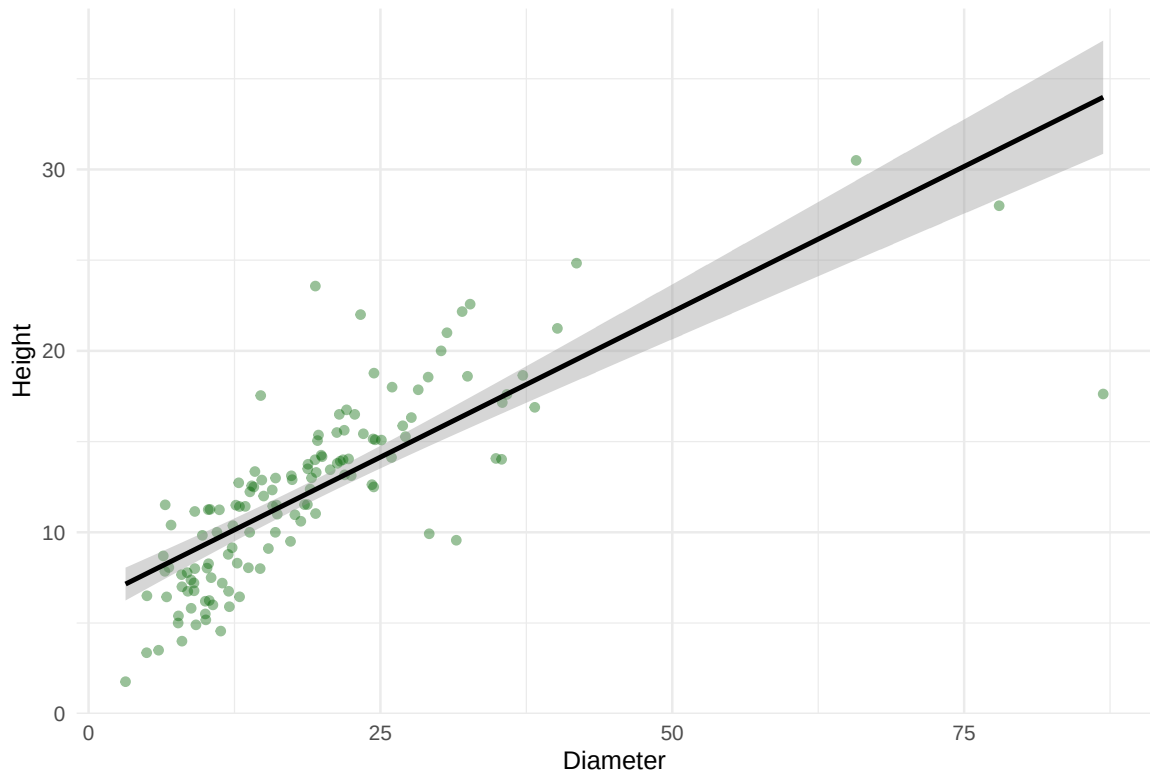
# Clean data
wd_clean <- wd_wide %>%
  filter(!is.na(DIAMETER), !is.na(HEIGHT)) %>%
  filter(is.finite(DIAMETER), is.finite(HEIGHT)) %>%
  filter(HEIGHT > 0, DIAMETER > 0) %>%          # remove zeros
  filter(DIAMETER < 150)                      # remove extreme outliers

# Overall model
model_all <- lm(HEIGHT ~ DIAMETER, data = wd_clean)

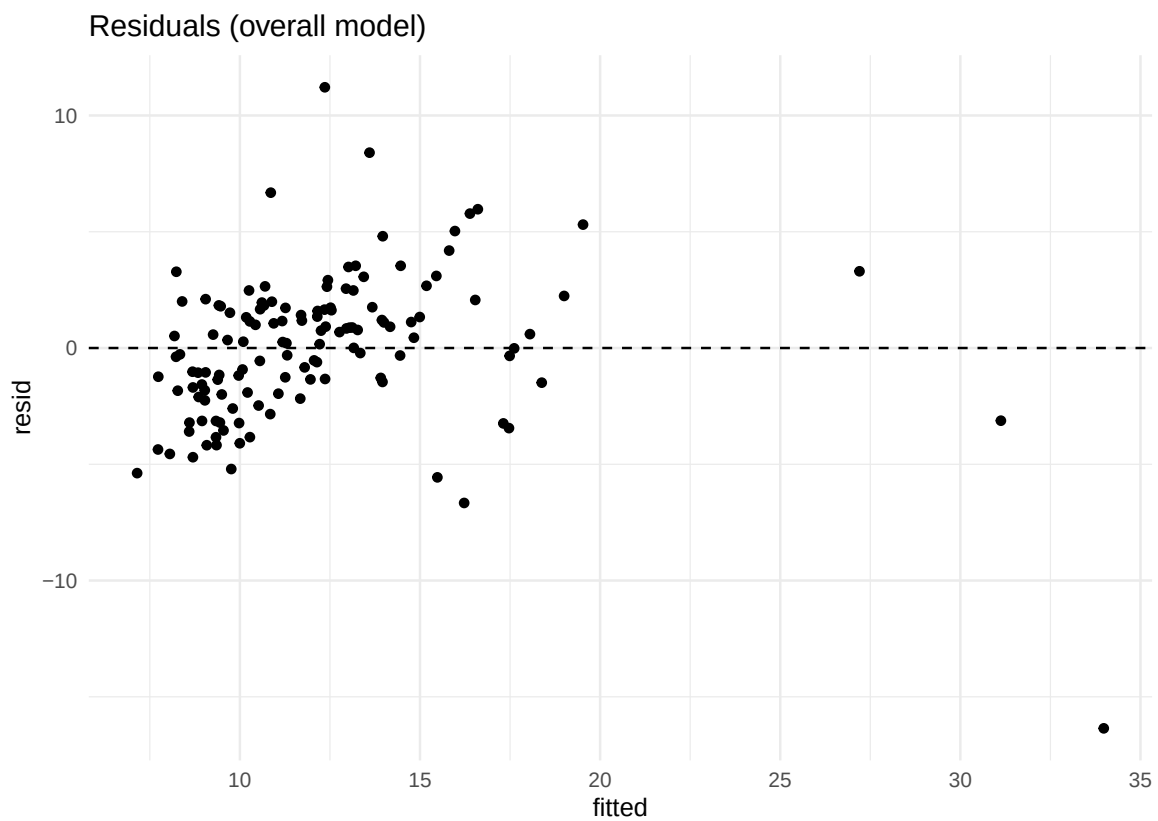
# Scatter plot
ggplot(wd_clean, aes(x = DIAMETER, y = HEIGHT)) +
  geom_point(alpha = 0.4, colour = "darkgreen") +
  geom_smooth(method = "lm", colour = "black", se = TRUE) +
  labs(title = "Overall model: HEIGHT vs DIAMETER",
       x = "Diameter",
       y = "Height") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```

Overall model: HEIGHT vs DIAMETER



```
# Residual plot
ggplot(data.frame(fitted = fitted(model_all), resid = resid(model_all)),
  aes(x = fitted, y = resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "Residuals (overall model)") +
  theme_minimal()
```



```
# Table
coef_all <- as.data.frame(summary(model_all)$coefficients)
```

```
coef_all <- round(coef_all, 4)
kable(coef_all, caption = "Overall model")
```

Table 2: Overall model

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.1328	0.5162	11.8796	0
DIAMETER	0.3204	0.0228	14.0361	0

```
# Site models
sites <- c("T05", "T06", "T08", "T09")

for(site in sites){

  data_site <- wd_clean %>% filter(SITECODE == site)

  model_site <- lm(HEIGHT ~ DIAMETER, data = data_site)

  # Scatter plot
  print(
    ggplot(data_site, aes(x = DIAMETER, y = HEIGHT)) +
    geom_point(alpha = 0.5, colour = "forestgreen") +
    geom_smooth(method = "lm", colour = "black", se = TRUE) +
    labs(title = paste("Site:", site),
         x = "Diameter",
         y = "Height") +
    theme_minimal()
  )

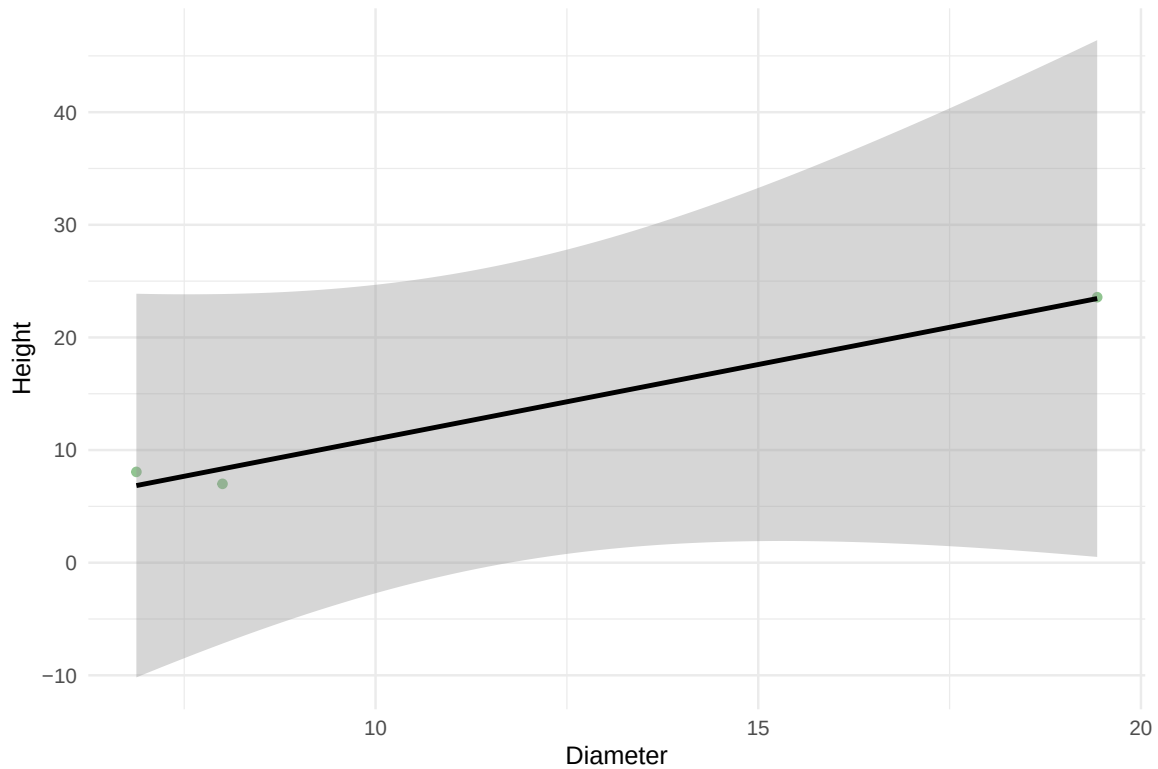
  # Residual plot
  print(
    ggplot(data.frame(fitted = fitted(model_site), resid = resid(model_site)),
         aes(x = fitted, y = resid)) +
    geom_point() +
    geom_hline(yintercept = 0, linetype = "dashed") +
    labs(title = paste("Residuals -", site)) +
    theme_minimal()
  )

  # Table
  coef_table <- as.data.frame(summary(model_site)$coefficients)
  coef_table <- round(coef_table, 4)

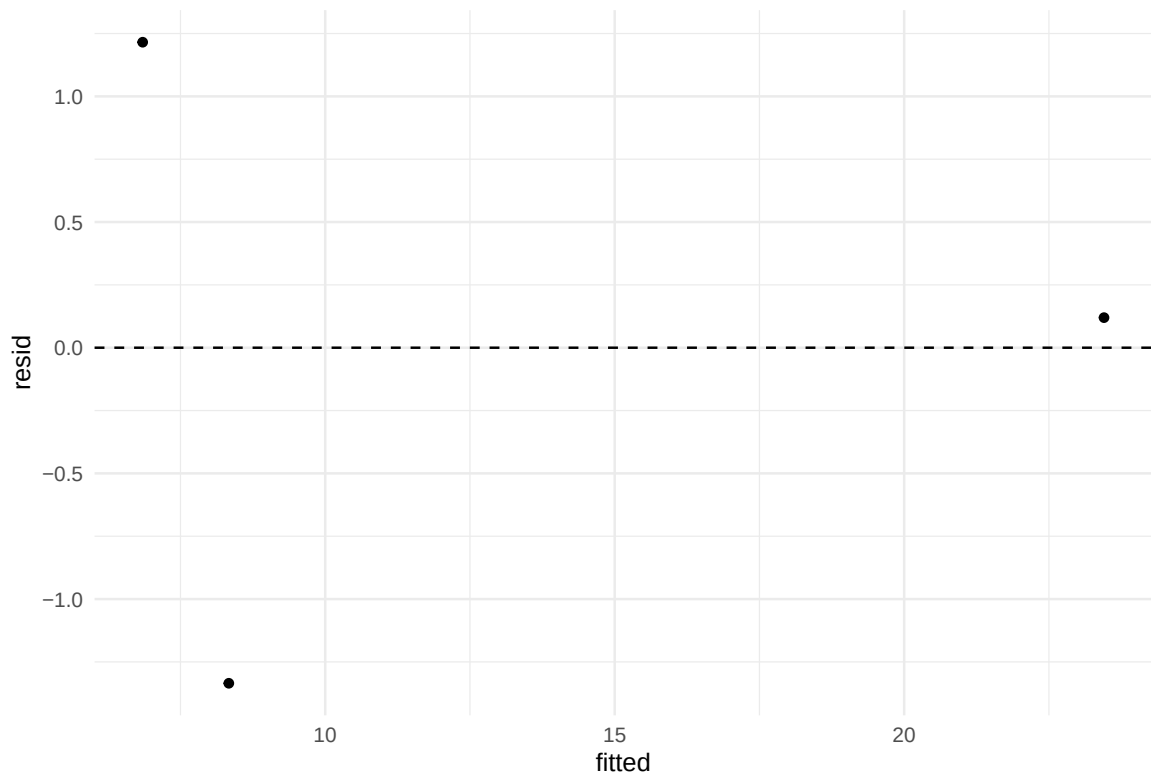
  print(kable(coef_table, caption = paste("Model for", site)))
}

## `geom_smooth()` using formula = 'y ~ x'
```

Site: T05

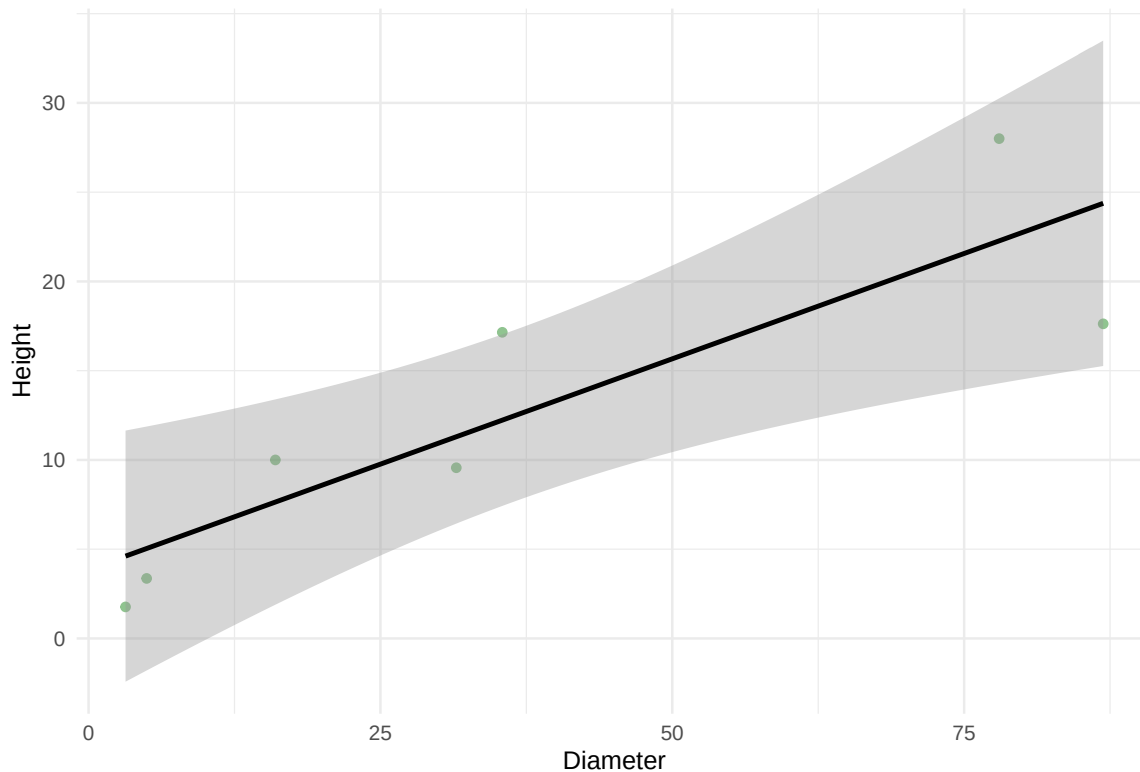


Residuals – T05

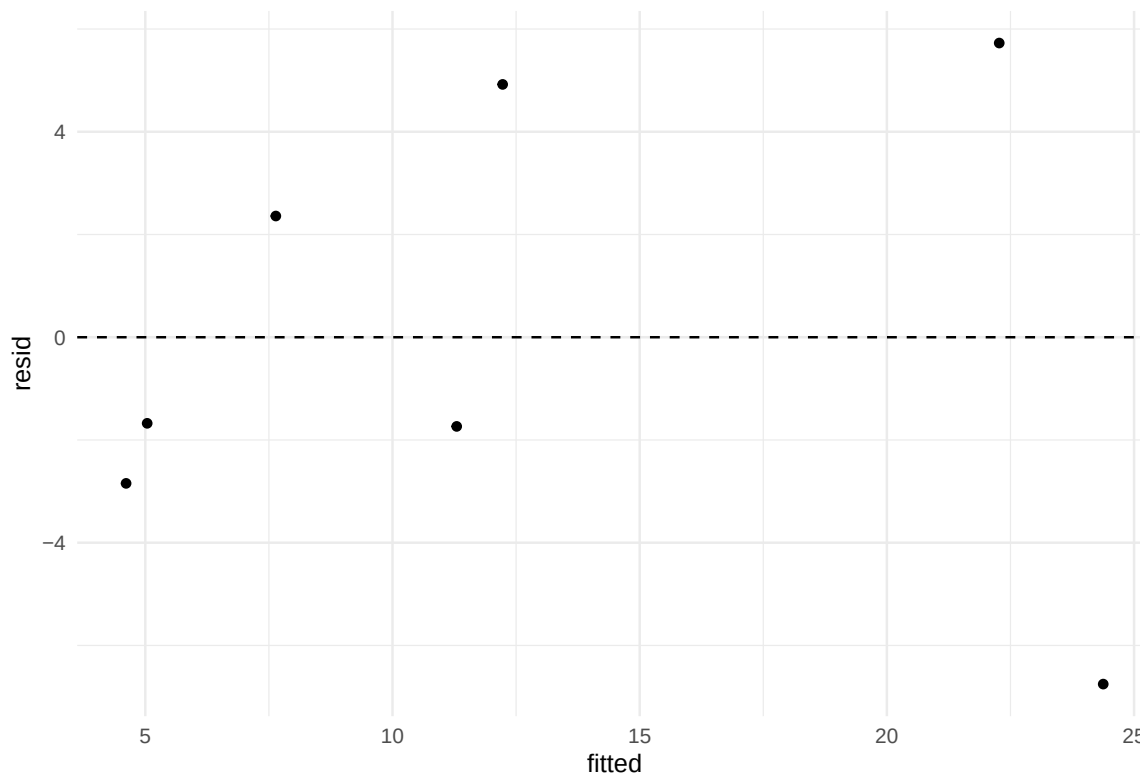


```
##
##
## Table: Model for T05
##
## |           | Estimate| Std. Error| t value| Pr(>|t|)|
## |:-----:|:-----:|:-----:|:-----:|:-----:|
## |(Intercept)| -2.2466| 2.3512| -0.9555| 0.5145|
## |DIAMETER   | 1.3227| 0.1842| 7.1806| 0.0881|
## `geom_smooth()` using formula = 'y ~ x'
```

Site: T06

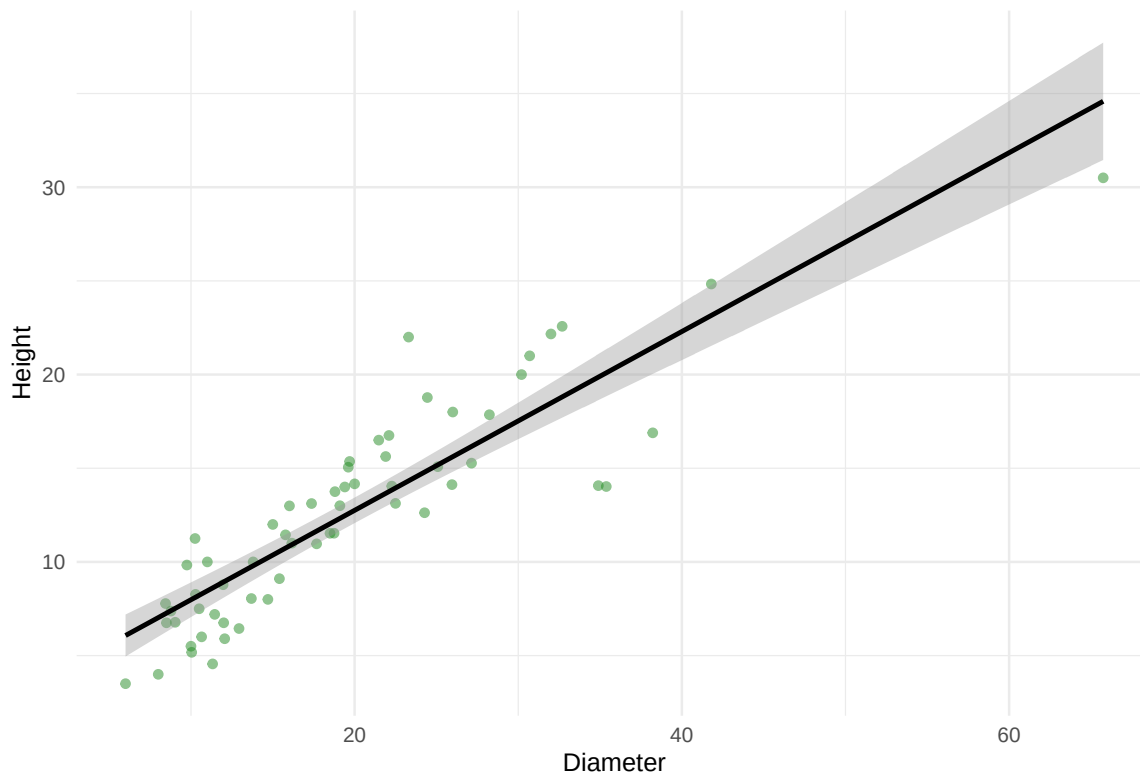


Residuals – T06

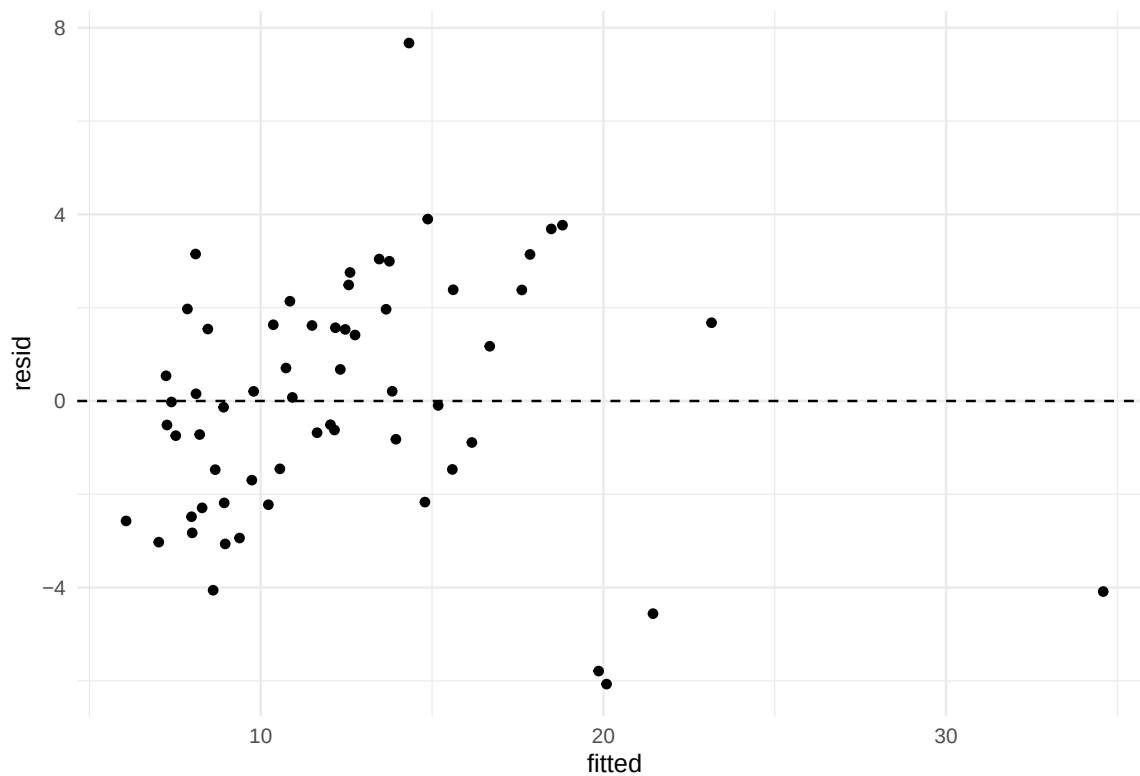


```
##
##
## Table: Model for T06
##
## |           | Estimate| Std. Error| t value| Pr(>|t|) |
## |-----|-----|-----|-----|-----|
## |(Intercept)|    3.864|    2.8777|  1.3427|  0.2371|
## |DIAMETER   |    0.236|    0.0599|  3.9426|  0.0109|
## `geom_smooth()` using formula = 'y ~ x'
```

Site: T08

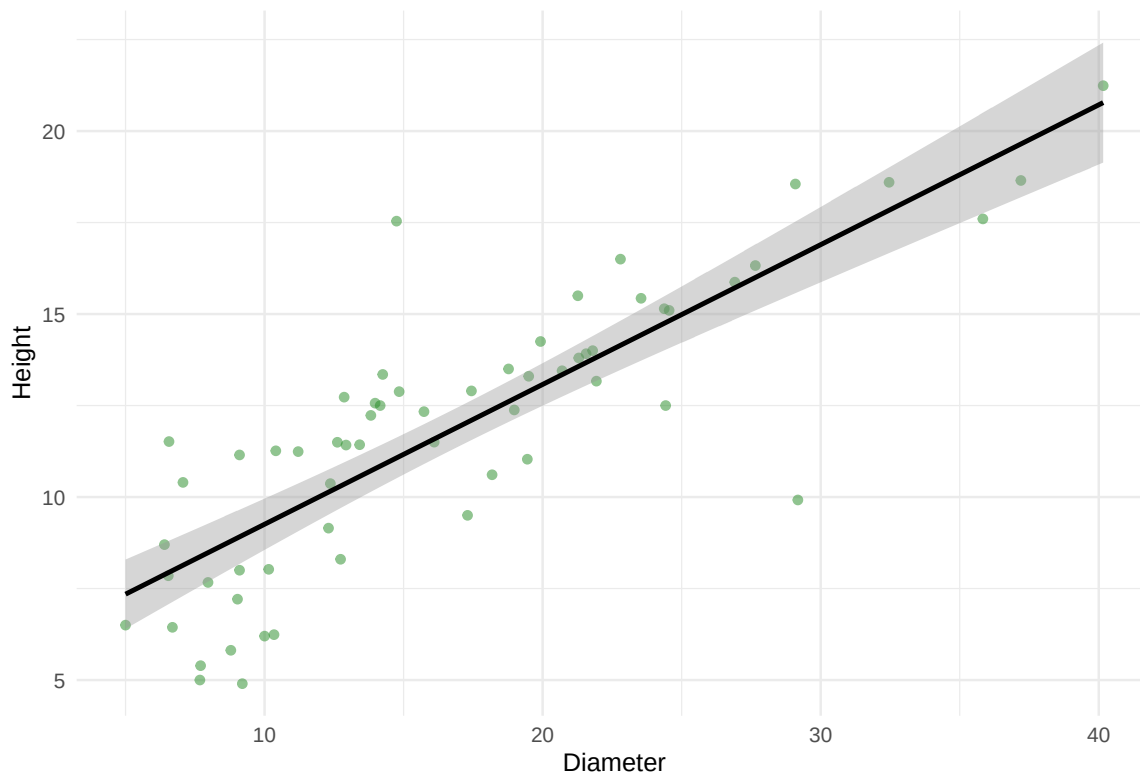


Residuals – T08

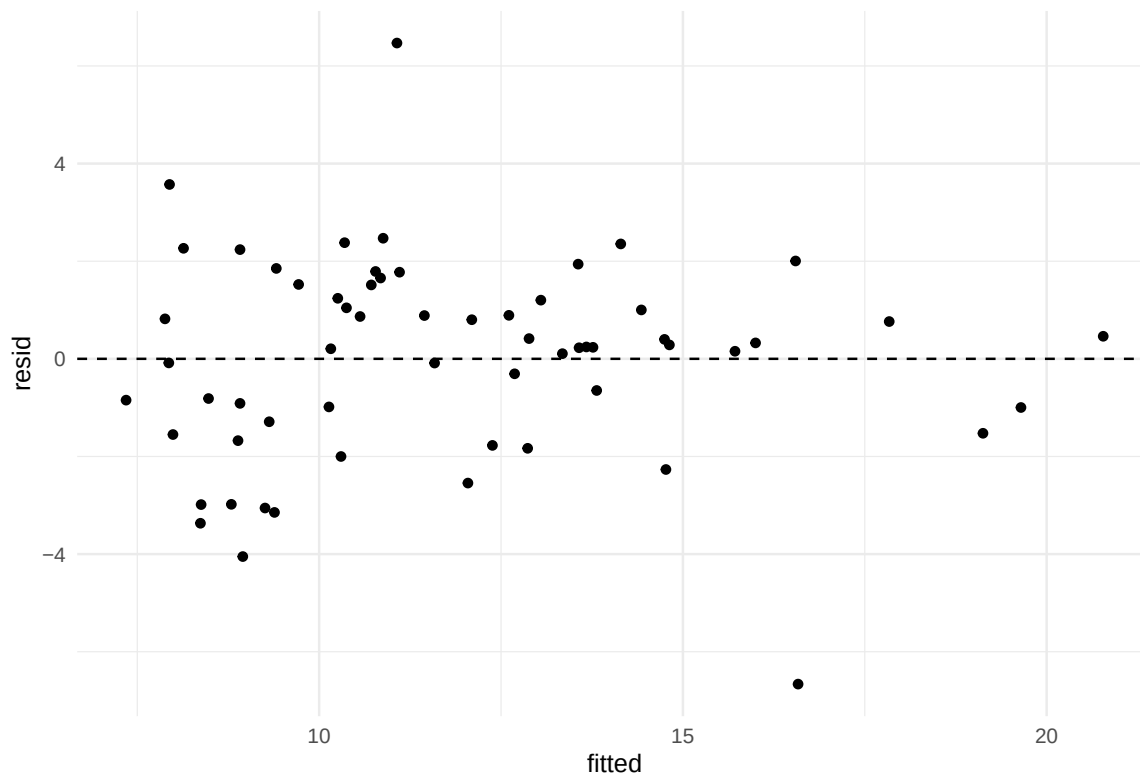


```
##
##
## Table: Model for T08
##
## |           | Estimate| Std. Error| t value| Pr(>|t|) |
## |-----|-----|-----|-----|-----|
## |(Intercept)|  3.2084|    0.7275|  4.4104|    0 |
## |DIAMETER   |  0.4772|    0.0330| 14.4548|    0 |
## `geom_smooth()` using formula = 'y ~ x'
```

Site: T09



Residuals - T09



```
##
##
## Table: Model for T09
##
## |           | Estimate| Std. Error| t value| Pr(>|t|) |
## |-----|-----|-----|-----|-----|
## |(Intercept)|  5.4361|    0.6159|   8.8256|      0|
## |DIAMETER   |  0.3820|    0.0330|  11.5620|      0|
```


d)

Different sites show features that make a simple linear model a bit shaky.

At T05, there are several very small or zero height values, which look unrealistic and create a strong cluster near zero. This leads to heteroscedasticity and weakens the linear relationship. Removing these values improves the speed and makes the linear trend clearer.

At T06, the main issue is very small sample size. With so few observations, the fitted regression line is unstable and has high variance, so a simple linear model is not really reliable here. In this case, it's better to acknowledge the limitation rather than overfit.

At T08, there are some large diameter values acting as outliers, which have high leverage and pull the regression line upwards. Filtering extreme values (e.g very large diameters) reduce their influence and results in a more representative fit for the majority of the data.

At T09, the spread of residuals increase with diameter, suggesting heteroscedasticity. A log transformation of the variables stabilises the variance and makes the relationship more linear. This improves model assumptions and produces a tighter fit.

Overall, these adjustments improve model fit by reducing bias from outliers, addressing heteroscedasticity, and ensuring the assumptions of linear regression (constant variance and linearity) are better met.

```
library(dplyr)
library(ggplot2)
library(knitr)

# Base cleaned data (from part c)
wd_clean <- wd_wide %>%
  filter(!is.na(DIAMETER), !is.na(HEIGHT)) %>%
  filter(is.finite(DIAMETER), is.finite(HEIGHT)) %>%
  filter(DIAMETER > 0, HEIGHT > 0)

sites <- c("T05", "T06", "T08", "T09")

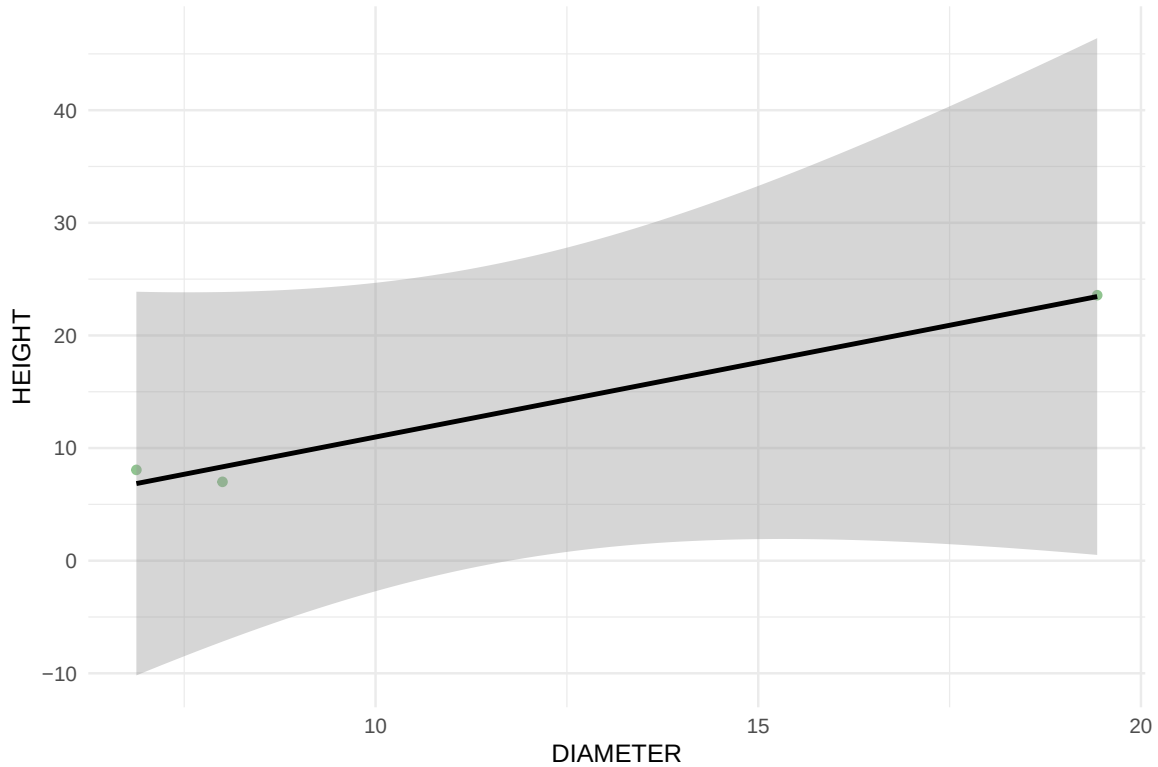
# T05: Remove zero / unrealistic heights
wd_T05 <- wd_clean %>%
  filter(SITECODE == "T05", HEIGHT > 1)

model_T05 <- lm(HEIGHT ~ DIAMETER, data = wd_T05)

ggplot(wd_T05, aes(DIAMETER, HEIGHT)) +
  geom_point(colour="forestgreen", alpha=0.5) +
  geom_smooth(method="lm", colour="black") +
  ggtitle("T05 (cleaned: removed near-zero heights)") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```

T05 (cleaned: removed near-zero heights)



```
kable(round(as.data.frame(summary(model_T05)$coefficients),4),
       caption="T05 refitted model")
```

Table 3: T05 refitted model

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.2466	2.3512	-0.9555	0.5145
DIAMETER	1.3227	0.1842	7.1806	0.0881

```
# T06: Keep but note limitation

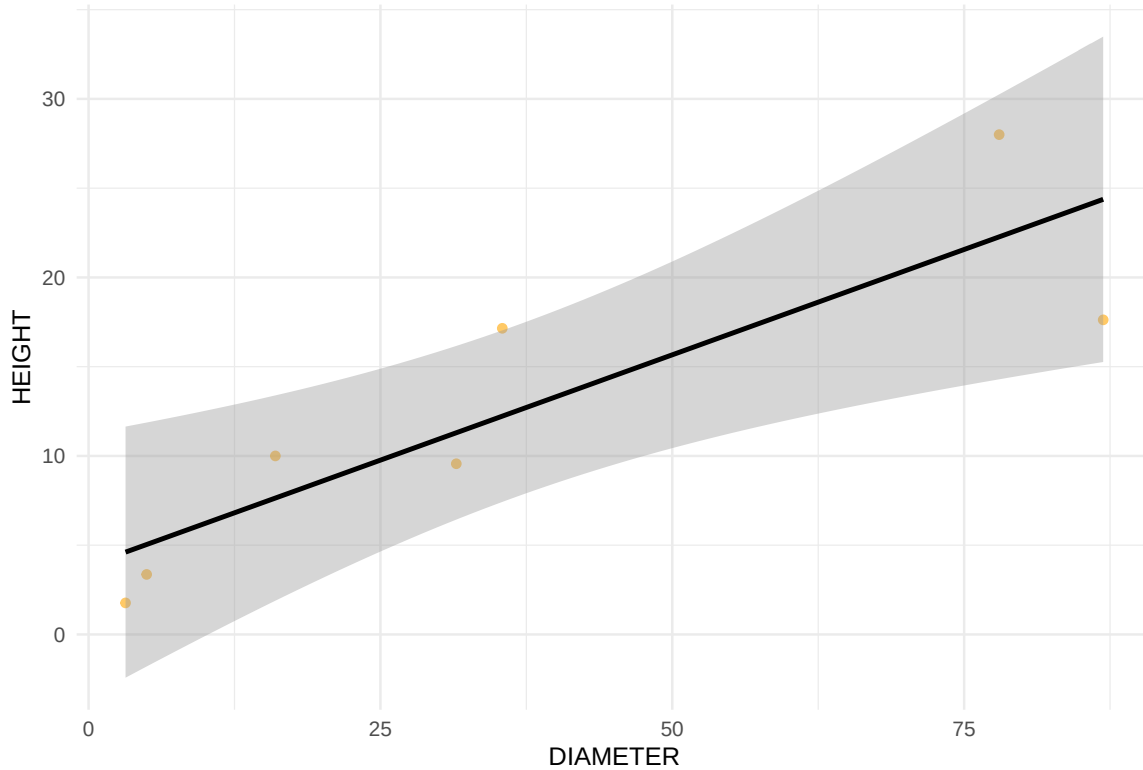
wd_T06 <- wd_clean %>% filter(SITECODE == "T06")

model_T06 <- lm(HEIGHT ~ DIAMETER, data = wd_T06)

ggplot(wd_T06, aes(DIAMETER, HEIGHT)) +
  geom_point(colour="orange", alpha=0.6) +
  geom_smooth(method="lm", colour="black") +
  ggtitle("T06 (small sample size)") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```

T06 (small sample size)



```
kable(round(as.data.frame(summary(model_T06)$coefficients),4),
       caption="T06 model (interpret cautiously)")
```

Table 4: T06 model (interpret cautiously)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.864	2.8777	1.3427	0.2371
DIAMETER	0.236	0.0599	3.9426	0.0109

```
# T08: Remove high-leverage outliers

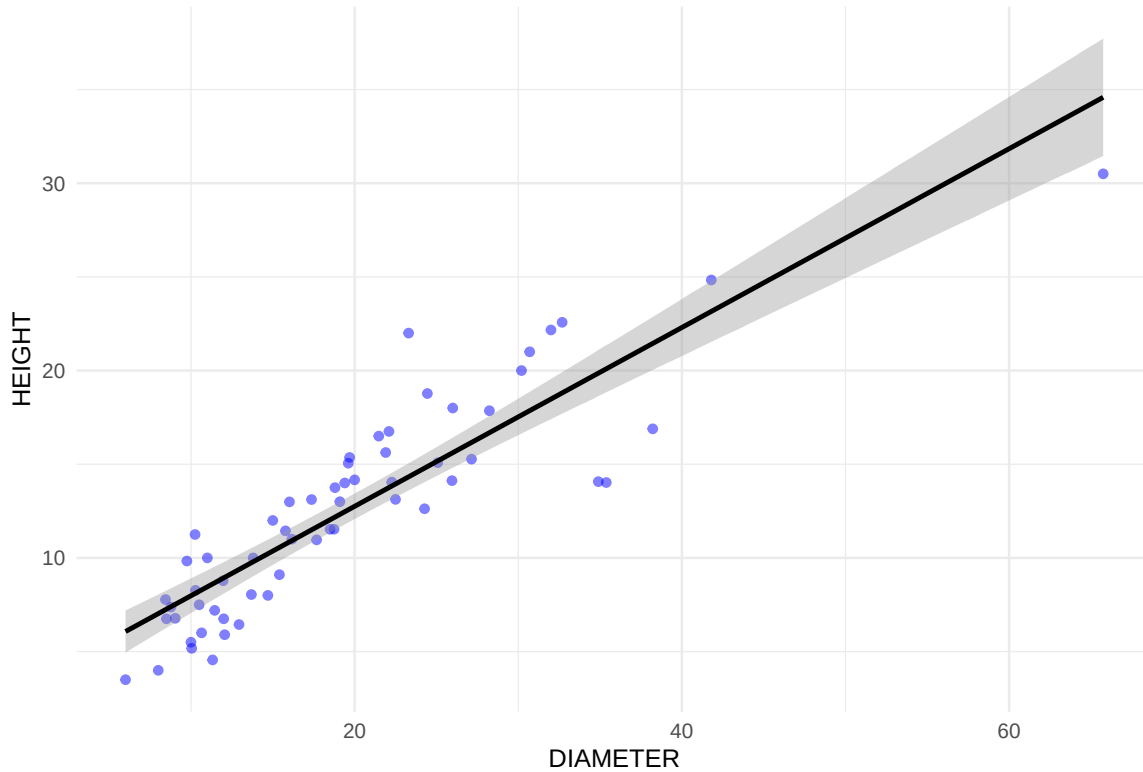
wd_T08 <- wd_clean %>%
  filter(SITECODE == "T08", DIAMETER < 150)

model_T08 <- lm(HEIGHT ~ DIAMETER, data = wd_T08)

ggplot(wd_T08, aes(DIAMETER, HEIGHT)) +
  geom_point(colour="blue", alpha=0.5) +
  geom_smooth(method="lm", colour="black") +
  ggtitle("T08 (outliers removed)") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```

T08 (outliers removed)



```
kable(round(as.data.frame(summary(model_T08)$coefficients),4),
      caption="T08 refitted model")
```

Table 5: T08 refitted model

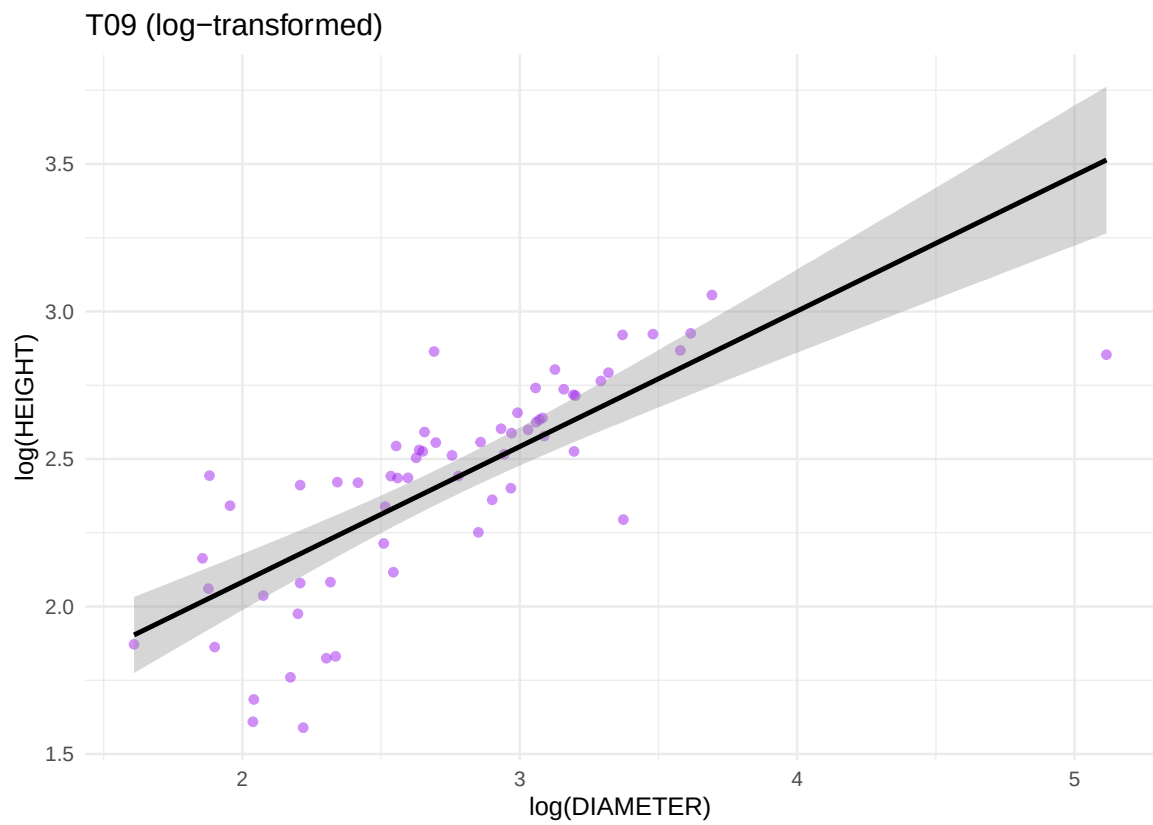
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.2084	0.7275	4.4104	0
DIAMETER	0.4772	0.0330	14.4548	0

```
# T09: Log transformation (fix heteroscedasticity)
wd_T09 <- wd_clean %>% filter(SITECODE == "T09")

model_T09 <- lm(log(HEIGHT) ~ log(DIAMETER), data = wd_T09)

ggplot(wd_T09, aes(log(DIAMETER), log(HEIGHT))) +
  geom_point(colour="purple", alpha=0.5) +
  geom_smooth(method="lm", colour="black") +
  ggtitle("T09 (log-transformed)") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



```
kable(round(as.data.frame(summary(model_T09)$coefficients),4),
       caption="T09 log-transformed model")
```

Table 6: T09 log-transformed model

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.1642	0.1422	8.1862	0
log(DIAMETER)	0.4593	0.0508	9.0392	0